

A Closed-Form Pressure Equation for the Accelerated $qx + 1$ Branching Process

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Abstract

We study the reverse tree of the accelerated $qx + 1$ map for odd $q \geq 3$ through a linear population functional. We prove an exact identity, $\rho_{\text{ann}}(\alpha) = q^{\alpha-1}/(2^\alpha - 1)$, for its annealed pressure, via an exact fibre-counting bijection. The nontrivial root of $\rho_{\text{ann}}(\alpha) = 1$ undergoes a structural transition at $q = 5$: it equals exactly $\alpha^* = 2$ at $q = 3$ and falls strictly below 1 for every $q \geq 5$. For $q = 3$ this is the classical Growth Exponent Conjecture of Kontorovich–Lagarias and Applegate–Lagarias. We show that the two roots of this equation always sit on opposite sides of a quenched/annealed freezing transition, in the sense of directed polymers in a random environment: the smaller root is always unfrozen, so the annealed identity transfers to the quenched counting function of a matching i.i.d. branching-random-walk model; the larger root is always frozen, so the pressure identity alone does not establish the tail index of the density martingale. We prove that the tilted q -adic measure is singular at every frozen root, by an entropy identity and a fractional-moment estimate, and we prove the tail index for the i.i.d. model by an implicit renewal theorem. For the coherent q -adic environment we prove martingale convergence and identify the limit with the density of the absolutely continuous component; absolute continuity and its tail index remain conjectural. A second, independent conjecture states the same tail index for the growth scale factor of the arithmetic reverse tree itself. At $q = 3$ the predicted index $\alpha^* = 2$ matches independent empirical measurements of that scale factor. At $q = 5$, the largest of three statistical tests on the scale factor, run at increasing sample size, gives evidence consistent with the conjectured index; an exact test of the martingale’s own tail remains inconclusive, and neither claim is proved for any $q \geq 5$.

1 Introduction

The Collatz, or $3x + 1$, conjecture generalizes naturally to a family of $qx + 1$ maps, q odd. This paper studies the reverse tree of the accelerated version of this map through a linear population functional G satisfying a natural fixed-point recursion, and derives, from an exact bijective counting argument, a closed-form equation for the annealed growth rate of the associated partition function. The equation’s roots organize the rest of the analysis: they govern a quenched/annealed freezing transition familiar from directed polymers in a random environment, and their relative position undergoes a structural change at $q = 5$ that reproduces, as a special case, a 30-year-old open conjecture of Kontorovich–Lagarias and Applegate–Lagarias for the classical Collatz map.

§2 fixes the reverse-tree construction. §3 proves the pressure identity and its consequences for the associated density martingale (absolute continuity criterion, singularity at frozen roots, Theorem 3.8). §4 develops the freezing transition. §5 states the structural transition at $q = 5$, separating what is proved for the matching stochastic model from what remains conjectural for the

actual arithmetic tree. §6 classifies the root $\alpha^* = 2$ at $q = 3$ against the multivariate smoothing-transform literature. §7 reports direct empirical support on real reverse trees.

2 Setup: the accelerated $qx + 1$ map and its reverse tree

Fix an odd integer $q \geq 3$ coprime to 2. The *accelerated $qx + 1$ map* is

$$T_q(n) = \frac{qn + 1}{2^{v_2(qn+1)}}, \quad n \text{ odd},$$

where v_2 is the 2-adic valuation. For $q = 3$ this is the standard accelerated Collatz map. The *reverse tree* rooted at an odd integer u coprime to q is built by, at each odd node u , adjoining a child $w_a := (2^a u - 1)/q$ for every $a \geq 1$ such that $2^a u \equiv 1 \pmod{q}$; such w_a is automatically an odd integer whenever it is an integer, since an odd numerator divided evenly by an odd denominator yields an odd quotient. Writing $d := \text{ord}_q(2)$ for the multiplicative order of 2 modulo q , admissibility of a for a given parent u depends only on $u \pmod{q}$: if some $a_0(u) \in \{1, \dots, d\}$ satisfies $2^{a_0(u)} u \equiv 1 \pmod{q}$, it is unique, and the admissible exponents are exactly $a \equiv a_0(u) \pmod{d}$. Since 1 is a unit mod q , $a_0(u)$ can exist only if u itself is a unit, so any u with $\gcd(u, q) > 1$, not merely $u \equiv 0 \pmod{q}$, is automatically *sterile* (no admissible children); among unit residues, $a_0(u)$ exists exactly when $u \pmod{q}$ lies in the subgroup $\langle 2 \rangle \leq (\mathbb{Z}/q\mathbb{Z})^\times$ generated by 2, and units outside $\langle 2 \rangle$ are sterile too. When 2 is a primitive root mod q (as for $q = 3, 5, 9$, all used numerically below), $\langle 2 \rangle$ is all of $(\mathbb{Z}/q\mathbb{Z})^\times$, so every unit residue is fertile and the sterile classes are exactly the non-units, i.e. the multiples of any prime factor of q : for prime q (so $q = 3, 5$) this is only $u \equiv 0$, recovering the familiar fact that multiples of 3 have trivial reverse subtrees; for $q = 9 = 3^2$ the sterile classes are $u \equiv 0, 3, 6 \pmod{9}$, already beyond the prime case. In general $d = \text{ord}_q(2)$ can also be a proper divisor of $\varphi(q)$ even among units: at $q = 7$, $d = 3 < \varphi(7) = 6$ and $\langle 2 \rangle = \{1, 2, 4\}$, so $u \equiv 3, 5, 6 \pmod{7}$ are sterile too, beyond the trivial $u \equiv 0$. This does not affect the pressure identity of Theorem 3.3 below (whose proof, via Lemma 3.2, works in the reverse direction, from an unconstrained exponent sequence to its unique root, and never invokes existence of $a_0(u)$ for every u), nor the numerical enumeration of §7, whose implementation already excludes exactly these classes; it is a correction to this structural description only.

We count distinct integer labels. Define

$$N_u(x) := \#\{n \leq x : n \text{ odd and } T_q^k(n) = u \text{ for some } k \geq 0\}.$$

This convention removes repetitions caused by cycles. In particular, $N_u(x) \leq \lceil x/2 \rceil$.

On this tree we study the linear population functional G satisfying the fixed-point recursion

$$G(u) = \sum_{a \equiv a_0(u) \pmod{d}} q \cdot 2^{-a} G(w_a), \quad (1)$$

which for $q = 3$ is exactly the martingale-type recursion underlying the endogeny-barrier line of investigation developed in a companion paper [1].

3 The pressure operator: an exact annealed identity

A first, natural attempt to organize the recursion (1) into a transfer operator is to track the parent's residue class modulo q alone. This attempt fails.

Example 3.1 (The naive mod- q automaton is not well defined). Take $q = 3$ (so $d = \text{ord}_3(2) = 2$) and $a = 2$, admissible for every parent $u \equiv 1 \pmod{3}$. Two parents $u = 1$ and $u = 7$, both $\equiv 1 \pmod{3}$, give $w_2(1) = (4 - 1)/3 = 1$ and $w_2(7) = (28 - 1)/3 = 9$, with $w_2(1) \equiv 1 \pmod{3}$ but $w_2(7) \equiv 0 \pmod{3}$: the child's residue class depends on $u \pmod{9}$, not merely on $u \pmod{3}$. In general, writing $u = u_0 + qk$ with $u_0 = u \pmod{q}$, one finds $w_a \pmod{q} = [(2^a u_0 - 1)/q \pmod{q}] + [2^a k \pmod{q}]$: an explicit dependence on $k = \lfloor u/q \rfloor \pmod{q}$. No finite modulus q^k repairs this: the same computation shows $w_a \pmod{q^k}$ depends on $u \pmod{q^{k+1}}$, one digit further than what mod- q^k knowledge of u supplies. There is therefore no well-defined finite-state transfer operator tracking a single generation of the recursion on residue classes of any fixed finite modulus.

What *is* well defined, and is the correct replacement for the naive picture of Example 3.1, is a bijection between an entire admissible fibre at one level of the digit tower and the full residue ring one level below.

Lemma 3.2 (Fibre bijection). *Fix $a \geq 1$ and $j \geq 1$. The map $v \mapsto w_a(v) := (2^a v - 1)/q$ is a bijection from $\{v \pmod{q^j} : 2^a v \equiv 1 \pmod{q}\}$ (an arithmetic progression of q^{j-1} residues mod q^j) onto $\mathbb{Z}/q^{j-1}\mathbb{Z}$.*

Proof. Well defined: $v \pmod{q^j}$ determines $2^a v - 1 \pmod{q^j}$, hence $w_a(v) \pmod{q^{j-1}}$. Injective: if $v \not\equiv v' \pmod{q^j}$ within the progression, write $v - v' = qt$ with $t \not\equiv 0 \pmod{q^{j-1}}$; then $w_a(v) - w_a(v') = 2^a t \not\equiv 0 \pmod{q^{j-1}}$ since 2 is a unit mod q^{j-1} . Surjectivity follows since the map is an injection between two finite sets of equal size q^{j-1} . $\square$

Iterating Lemma 3.2 along a depth- k path $(a_1, \dots, a_k)$ shows every sequence $(a_1, \dots, a_k) \in (\mathbb{Z}_{\geq 1})^k$, with no admissibility restriction on the individual a_i , is admissible for *exactly one* root residue $u_0 \pmod{q^k}$, namely $u_0 \equiv \sum_{i=1}^k q^{i-1} 2^{-S_i} \pmod{q^k}$, $S_i := a_1 + \dots + a_i$. Define the depth- k partition function rooted at u_0 ,

$$Z_k(\alpha; u_0) := \sum_{\substack{(a_1, \dots, a_k) \text{ admissible} \\ \text{from } u_0}} \prod_{i=1}^k (q \cdot 2^{-a_i})^\alpha.$$

Theorem 3.3 (Exact annealed pressure identity). *For every odd $q \geq 3$, every $k \geq 1$, and every $\alpha > 0$,*

$$\sum_{u_0 \in \mathbb{Z}/q^k\mathbb{Z}} Z_k(\alpha; u_0) = \left(\frac{q^\alpha}{2^\alpha - 1} \right)^k, \quad (2)$$

so the average of $Z_k(\alpha; \cdot)$ over all q^k root residues equals exactly $(q^{\alpha-1}/(2^\alpha - 1))^k$; the associated annealed pressure is

$$\rho_{\text{ann}}(\alpha) = \frac{q^{\alpha-1}}{2^\alpha - 1}, \quad (3)$$

giving the pressure equation at $\rho_{\text{ann}} = 1$:

$$q^{\alpha-1} = 2^\alpha - 1. \quad (4)$$

Proof. By Lemma 3.2 (iterated), the admissible depth- k sequences rooted at the q^k distinct residues u_0 partition the full, unconstrained set $(\mathbb{Z}_{\geq 1})^k$ with no overlap and no omission. Summing $\prod_i (q 2^{-a_i})^\alpha$ over all sequences in $(\mathbb{Z}_{\geq 1})^k$ therefore factors as $(\sum_{a=1}^\infty (q 2^{-a})^\alpha)^k = (q^\alpha / (2^\alpha - 1))^k$, which is exactly the left-hand side of (2) summed over the partition. $\square$

We verified (2) independently by direct enumeration for $q \in \{3, 5, 7, 9\}$, $k \in \{1, 2, 3\}$, and $\alpha \in \{0.26, 0.5, 1, 1.37, 2\}$, matching to the truncation error of the (rapidly convergent) tail of the sum over a .

Remark 3.4 (Mass conservation as a corollary). The right-hand side of (2) evaluates to exactly $1^k = 1$ at $\alpha = 1$ for every q (mass conservation of the branching process). Thus $\alpha = 1$ is always a root of (4).

Lemma 3.5 (Log-convexity of the pressure function). *For every odd $q \geq 3$, $P(\alpha) := \log \rho_{\text{ann}}(\alpha)$ is strictly convex on $\alpha > 0$. Consequently equation (4) has at most two positive roots, and if it has exactly two, $\alpha_- < \alpha_+$, then $P'(\alpha_-) < 0 < P'(\alpha_+)$.*

Proof. From (3), $P(\alpha) = (\alpha - 1) \log q - \log(2^\alpha - 1)$, so

$$P''(\alpha) = \frac{\log^2 2 \cdot 2^\alpha}{(2^\alpha - 1)^2} > 0$$

for every $\alpha > 0$: strict convexity. A strictly convex function equals any given value at most twice, and, at two such roots, must be decreasing through the first and increasing through the second (on either side of its unique minimum), giving $P'(\alpha_-) < 0 < P'(\alpha_+)$. $\square$

At a pressure root α , the numbers $p_\alpha(a) := q^{\alpha-1} 2^{-\alpha a}$, $a \geq 1$, sum to one, so they define a probability law on branch exponents. For independent digits $a_1, a_2, \dots$ drawn from p_α , the partial sums $F_k := \sum_{i=1}^k q^{i-1} 2^{-S_i} \pmod{q^k}$, $S_i := a_1 + \dots + a_i$, form a projectively consistent family (Lemma 3.2 gives $F_{k+1} \equiv F_k \pmod{q^k}$), hence converge to a $\mathbb{Z}_q$ -valued limit; write μ_α for its law, the *tilted Syracuse measure* at α .

Theorem 3.6 (q -adic density martingale). *Let α be either positive root of (4), and let U be Haar-uniform on $\mathbb{Z}_q$. Then*

$$M_k^{(\alpha)}(U) := Z_k(\alpha; U \bmod q^k)$$

is a nonnegative martingale of mean one for the filtration generated by $U \bmod q^k$. Consequently M_k converges almost surely to a finite limit $W_{q,\alpha}$. Writing $\mu_\alpha = f_{q,\alpha} m_q + \mu_s$ for the Lebesgue decomposition of the tilted Syracuse measure above with respect to Haar measure m_q , then

$$W_{q,\alpha} = f_{q,\alpha}(U) \quad m_q\text{-almost surely.}$$

In particular, $W_{q,\alpha}$ is nonzero with positive probability exactly when μ_α has a nonzero absolutely continuous component, and $\mathbb{E}W_{q,\alpha} = 1$ exactly when $\mu_\alpha \ll m_q$.

Proof. With F_k and p_α as above, the fibre identity gives

$$Z_k(\alpha; u) = q^k \Pr(F_k = u).$$

Conditional on $U \equiv u \pmod{q^k}$,

$$\mathbb{E}[M_{k+1} \mid U \bmod q^k = u] = \frac{1}{q} \sum_{v \equiv u \pmod{q^k}} q^{k+1} \Pr(F_{k+1} = v) = q^k \Pr(F_k = u) = M_k(u).$$

The nonnegative martingale convergence theorem gives the limit. The functions M_k are the Radon–Nikodym derivatives of the restrictions of μ_α to the cylinder sigma-algebras relative to the corresponding restrictions of m_q . Martingale differentiation of the Lebesgue decomposition identifies their almost-sure limit with $d(\mu_\alpha)_{\text{ac}}/dm_q$. Its integral is the total mass of the absolutely continuous part, which proves the final assertions. $\square$

Theorem 3.7 (L^p collision criterion). *Let α be a pressure root, let $p > 1$, let $\mu_{\alpha,k}$ be the law of F_k modulo q^k , and let $M_k^{(\alpha)} = q^k \mu_{\alpha,k}$ be its density relative to uniform measure on $\mathbb{Z}/q^k\mathbb{Z}$. Then*

$$\|M_k^{(\alpha)}\|_p^p = q^{k(p-1)} \sum_{u \bmod q^k} \mu_{\alpha,k}(u)^p. \quad (5)$$

The projective measure μ_α has a density in $L^p(\mathbb{Z}_q)$ if and only if the right-hand side is bounded uniformly in k . If p is an integer, the sum in (5) is the probability that p independent tilted Syracuse sums agree modulo q^k .

Proof. Uniform measure assigns mass q^{-k} to each residue, so

$$\|M_k^{(\alpha)}\|_p^p = q^{-k} \sum_u (q^k \mu_{\alpha,k}(u))^p,$$

which is (5). For integer p , expanding the probability of equality of p independent copies gives the collision interpretation.

If the quantities in (5) are uniformly bounded, the martingale $M_k^{(\alpha)}$ converges in L^p . Its limit has integral one and is the density of the projective measure on every cylinder, so μ_α is absolutely continuous with an L^p density. Conversely, if $d\mu_\alpha = f dm_q$ with $f \in L^p$, then $M_k^{(\alpha)} = \mathbb{E}[f \mid U \bmod q^k]$. Conditional expectation is an L^p contraction, which gives the uniform bound. $\square$

For $q = 3$ and $\alpha = 1$, Theorem 3.7 at $p = 2$ gives exactly the collision statistic used in the finite-level L^2 growth measurement of the endogeny-barrier companion paper [1]. Thus that L^2 sufficient condition is one member of a complete family of weighted collision criteria (finite-level spectrum computed in the accompanying repository). These are averaged concentration conditions; they do not control the smallest cylinder mass required by a worst-residue covering problem.

Theorem 3.8 (Singularity at the frozen root). *Let $\alpha = \alpha_+(q)$, the larger of the two positive roots of (4) (§4). Then*

$$M_k^{(\alpha)}(U) \longrightarrow 0 \quad m_q\text{-almost surely,}$$

and the tilted Syracuse measure μ_α is singular with respect to Haar measure.

Proof. Write $r = 2^{-\alpha}$. The digit law is $p_\alpha(a) = (1-r)r^{a-1}$, so its entropy is

$$H(p_\alpha) = -\log(1-r) - \frac{r}{1-r} \log r.$$

At a pressure root, direct differentiation gives

$$H(p_\alpha) - \log q = -\alpha P'(\alpha) = s(\alpha).$$

The larger root is frozen, hence $H(p_\alpha) < \log q$ (see §4). For $0 < t < 1$, subadditivity of x^t and the representation $M_k^{(\alpha)}(u) = q^k \sum_{F_k=u} \prod_i p_\alpha(a_i)$ give

$$\mathbb{E}_{m_q}[(M_k^{(\alpha)})^t] \leq \left(q^{t-1} \sum_{a \geq 1} p_\alpha(a)^t \right)^k.$$

The logarithmic derivative of the factor in parentheses at $t = 1$ is $\log q - H(p_\alpha) > 0$. It is therefore less than one for some $t < 1$ sufficiently close to one. The fractional moment tends to zero, so the almost-sure martingale limit is zero. Theorem 3.6 then identifies the absolutely continuous component as zero. $\square$

4 Quenched vs. annealed: a freezing transition

Theorem 3.3 is a statement about the *average* of $Z_k(\alpha; \cdot)$ over root residues; it is not, by itself, a statement about $Z_k(\alpha; u_0)^{1/k}$ for a single, fixed root u_0 , which is what actually governs the population martingale $G(v)$ for one specific integer v . Whether the two coincide is exactly the quenched-versus-annealed question familiar from directed polymers in a random environment and from Derrida's random energy model, governed by a standard freezing criterion. Writing $P(\alpha) := \log \rho_{\text{ann}}(\alpha)$ for the annealed log-pressure and $s(\alpha) := P(\alpha) - \alpha P'(\alpha)$ for the associated entropy, quenched and annealed rates coincide where $s(\alpha) > 0$ (an *unfrozen* phase) and diverge where $s(\alpha) < 0$ (a *frozen* phase, in which rare, low-entropy paths dominate the quenched rate instead of the bulk of the ensemble).

Proposition 4.1 (The larger root is always frozen). *For every odd $q \geq 3$, writing $\alpha_- < \alpha_+$ for the two positive roots of $\rho_{\text{ann}}(\alpha) = 1$: $s(\alpha_-) > 0$ (unfrozen) and $s(\alpha_+) < 0$ (frozen), unconditionally.*

Proof. At any root, $P(\alpha) = 0$, so $s(\alpha) = -\alpha P'(\alpha)$. Lemma 3.5 gives $P'(\alpha_-) < 0 < P'(\alpha_+)$ directly. Hence $s(\alpha_-) = -\alpha_- P'(\alpha_-) > 0$ and $s(\alpha_+) = -\alpha_+ P'(\alpha_+) < 0$. $\square$

For $q \geq 5$, where $\alpha_+ = 1$ always, this has an exact closed form: at $\alpha = 1$, $2^\alpha - 1 = 1$ so $\log(2^\alpha - 1) = 0$, and $s(1) = 2 \log 2 - \log q = \log(4/q)$, which is negative exactly when $q > 4$. So the freezing of the trivial root for every odd $q \geq 5$ is not a numerical coincidence but the same sign condition as the mean-drift remark below ($\log q - 2 \log 2 > 0$): the drift that makes orbits expand on average is, up to a minus sign, exactly the entropy deficit that freezes $\alpha_+ = 1$.

Since $s'(\alpha) = P'(\alpha) - P'(\alpha) - \alpha P''(\alpha) = -\alpha P''(\alpha) < 0$ for every $\alpha > 0$ by Lemma 3.5, s is strictly decreasing throughout, so its root $\alpha_c(q)$ is unique wherever it exists; existence, and that it lies strictly between $\alpha_-(q)$ and $\alpha_+(q)$, follows from Proposition 4.1 by the intermediate value theorem. We computed $\alpha_c(q)$ directly:

q	$\alpha_c(q)$
3	1.355
5	0.794
7	0.564
9	0.437

confirming $\alpha_-(q) < \alpha_c(q) < \alpha_+(q)$ in every case, as Proposition 4.1 requires. We also verified numerically, with an independent memoised recursion computing $Z_k(\alpha; u_0)$ exactly at a fixed (not averaged) root, that the freezing reflects a real divergence of quenched from annealed growth: at $q = 3$, $u_0 = 1$ (a period-one cycle of the residue dynamics, since $w_2(1) = 1$ exactly), $Z_k(2; 1)^{1/k}$ shows an oscillation with geometric-mean ratio well below the annealed rate $\rho_{\text{ann}}(2) = 1$; at $q = 5$, $Z_k(1; u_0)^{1/k}$ at the roots $u_0 \in \{1, 2, 3, 4\}$ likewise fails to track $\rho_{\text{ann}}(1) = 1$. This check has a limitation: every one of these roots lies on or feeds directly into the smallest-admissible-branch map's own trivial cycle ($1 \leftrightarrow 3$ for $q = 5$, exactly as $u_0 = 1$ does for $q = 3$), so the divergence observed here has a deterministic-recurrence explanation available in addition to the entropy mechanism of Proposition 4.1, and does not by itself sample the generic (non-periodic) environment that the quenched statement is really about. We therefore report this as an illustration of freezing at a special, easily-computed class of roots, not as a confirmation at a generic one; §7 below carries the corresponding generic-environment evidence, a handful of roots at $q = 5, 7$ sampled well away from any known short cycle.

The pressure identity transfers to the counting exponent of the i.i.d. model at the unfrozen root $\alpha_-(q)$. It does not by itself prove a tail law. Theorem 5.6 obtains the i.i.d. tail by implicit

renewal, while Conjecture 5.5 states the separate claim for a coherent Haar-random q -adic root environment.

5 The structural transition at $q = 5$

Besides the trivial root $\alpha = 1$, equation (4) has exactly one further positive root $\alpha^*(q)$. By Lemma 3.5, $\rho_{\text{ann}}(\alpha) = 1$ has at most two positive roots; for existence, $\rho_{\text{ann}}(\alpha) \rightarrow \infty$ both as $\alpha \rightarrow 0^+$ (since $2^\alpha - 1 \rightarrow 0$ linearly while $q^{\alpha-1} \rightarrow 1/q$) and as $\alpha \rightarrow \infty$ (since $q > 2$), so $P := \log \rho_{\text{ann}} \rightarrow \infty$ at both ends, and $P'(1) = \log(q/4) \neq 0$ for every odd q (§4) places the strictly convex function's unique minimum away from $\alpha = 1$, forcing a second root on the side of 1 that the sign of $P'(1)$ selects. We solved for $\alpha^*(q)$ independently by bisection:

q	second root $\alpha^*(q)$	interpretation
3	2.000000	exact
5	0.650919	
7	0.373501	
9	0.258108	

At $q = 3$ the second root is exactly $\alpha^* = 2$, matching the universal tail exponent for the classical Collatz reverse tree established by independent methods (Hill estimator on real-tree samples, and extreme value theory on block maxima) in earlier stages of this line of work. For $q \geq 5$, however, the second root falls strictly *below* 1, a qualitative structural transition. Writing $\alpha_- < \alpha_+$ for the two roots, we have $\{\alpha_-, \alpha_+\} = \{1, 2\}$ at $q = 3$ but $\{\alpha_-, \alpha_+\} = \{\alpha^*(q), 1\}$ for $q \geq 5$: the trivial root becomes the *larger* of the two.

The i.i.d. model. Idealize the reverse tree by declaring each candidate exponent $a \geq 1$ independently fertile with probability $1/q$; a fertile a produces a child at multiplicative displacement $2^a/q$ from its parent, matching the arithmetic tree's own transition rule (§2) one branch at a time rather than only in aggregate. Writing $S_v \in \mathbb{R}$ for the cumulative log-displacement of an individual v from the root, and $N(x) := \#\{v : S_v \leq \log x\}$ for the number of individuals of any generation reaching size at most x , the many-to-one formula gives, for $\theta > 0$,

$$\mathbb{E} \left[\sum_{|v|=n} e^{-\theta S_v} \right] = \left(\mathbb{E} \left[\sum_{a \geq 1} \mathbf{1}[\text{fertile}] (q/2^a)^\theta \right] \right)^n = \left(\frac{1}{q} \sum_{a \geq 1} (q/2^a)^\theta \right)^n = \rho_{\text{ann}}(\theta)^n,$$

so the model's per-generation growth rate is governed by exactly the same pressure function as Theorem 3.3.

Theorem 5.1 (Structural transition: exponent, i.i.d. model). *For the i.i.d. model above, $\log N(x)/\log x \rightarrow \alpha_-(q)$ almost surely, for every odd $q \geq 3$: exponent 1 (i.e. $N(x) = x^{1+o(1)}$) at $q = 3$; exponent $\alpha_-(q) < 1$ for every $q \geq 5$.*

Proof. Upper bound. Fix $\theta \in (\alpha_-(q), \alpha_+(q))$, where $\rho_{\text{ann}}(\theta) < 1$ by the log-convexity argument above (a convex function vanishing at two points $\alpha_- < \alpha_+$ is non-positive in between). By Markov's inequality applied to the many-to-one identity, $\mathbb{E}[\#\{|v| = n : S_v \leq y\}] \leq e^{\theta y} \rho_{\text{ann}}(\theta)^n$, so summing over all generations, $\mathbb{E}[N(e^y)] \leq e^{\theta y}/(1 - \rho_{\text{ann}}(\theta))$. A second Markov step gives $\Pr(N(x) > x^{\theta+\varepsilon}) \leq x^{-\varepsilon}/(1 - \rho_{\text{ann}}(\theta))$ for any $\varepsilon > 0$; this is summable along $x = 2^k$, so Borel–Cantelli and monotonicity of N give $\limsup_x \log N(x)/\log x \leq \theta + \varepsilon$ almost surely. Letting $\theta \downarrow \alpha_-(q)$ and $\varepsilon \downarrow 0$ gives $\limsup \leq \alpha_-(q)$.

Lower bound. This direction, matching the upper bound exponent almost surely on the survival event, is the content of Kontorovich–Lagarias’s Theorem 8.10 [3], proved there for the single-type $5x + 1$ branching random walk by a large-deviations argument adapted from Lagarias–Weiss (the same Chernoff computation as above, run in reverse to locate the generation k at which the bulk of the count accumulates, together with a second-moment estimate showing this generation’s contribution is not a rare event). Their argument uses only the branching structure’s pressure function, the independence of fertility across nodes, and survival on the large-deviation event with positive probability, the last of which is exactly what unfreezing at $\alpha_-(q)$ (Proposition 4.1) supplies; with that in hand the argument transfers without change, and we do not repeat the second-moment computation here. $\square$

Proposition 5.2 (Finer asymptotic, conditional). *The counting function conjecturally satisfies the sharper asymptotic $N(x) \sim W \cdot x^{\alpha_-(q)}$, not merely Theorem 5.1’s exponent statement, holding almost surely and uniformly, including at $q = 3$, for an almost surely positive, non-degenerate random scale factor W (not a deterministic constant at any q , since the underlying reproduction law has nonzero variance). This is a strictly stronger claim than Theorem 5.1: it pins down the constant in front of $x^{\alpha_-(q)}$, which needs machinery beyond the large-deviations argument used above. A directed literature search found the neighboring machinery but not this exact statement ready-made: the classical Crump–Mode–Jagers renewal theorem [10] is stated for one-sided $([0, \infty))$ reproduction point processes, while our branching displacements include negative values (e.g. $2/3 < 1$ at $q = 3$, since a child can be numerically smaller than its parent); the branching-random-walk martingale theory that is natively two-sided [9] gives convergence generation-by-generation, not a renewal law summed over all generations; and the implicit renewal theory for weighted trees with general (two-sided) weights [11, 12, 13] resolves the tail of the fixed-point equation for W itself, not the cumulative counting law for $N(x)$, a related but distinct question. We expect the present statement to follow from combining this machinery (spine decomposition and exponential tilting for a two-sided one-dimensional random walk, exactly as already used for the tail-of- W results just cited) but we have not carried out that adaptation, nor located it already written in this exact form. A very recent preprint [14] extends Biggins-type martingale convergence to general type spaces and non-geometric weights, but at the level of individual generations, the same scope as [9] above; it contains no renewal or Crump–Mode–Jagers machinery and does not address the cumulative, summed-over-generations statement needed here. The non-lattice hypothesis that any such extension of [10] would need holds here, and it is exactly the irrationality of $\log q / \log 2$ that secures it. Although each single-step displacement $-\log q + a \log 2$ ($a \in \mathbb{Z}_{\geq 1}$) lies in one coset of the lattice $\log(2)\mathbb{Z}$, containment in a coset is not arithmeticity: the smallest closed subgroup of $\mathbb{R}$ generated by these displacements is $\log(q)\mathbb{Z} + \log(2)\mathbb{Z}$, which is dense in $\mathbb{R}$ for every odd $q \geq 3$ because $\log q / \log 2 \notin \mathbb{Q}$. Equivalently, the characteristic function of the displacement equals 1 only at the origin, so the renewal/CMJ asymptotic carries no log-periodic (period $\log 2$) modulation. Were $\log q / \log 2$ rational the branch positions would collapse onto a lattice and a periodic correction would appear.*

Conjecture 5.3 (Structural transition: exponent, arithmetic tree). *For every admissible root u , the actual reverse tree of T_q satisfies*

$$N_u(x) = x^{\alpha_-(q)+o(1)}.$$

Thus the exponent is 1 for $q = 3$ and is strictly below 1 for every odd $q \geq 5$. At $q = 3$ this is the Growth Exponent Conjecture of Kontorovich–Lagarias [3] and Applegate–Lagarias [4, 5] for the classical Collatz reverse tree, namely $\pi_a(x) = x^{1+o(1)}$ for their analogue of $N_u(x)$ above (the count, up to x , of odd integers reaching a fixed root a under repeated reverse iteration), open since 1995,

with the best unconditional partial result remaining the one-sided bound $\pi_a(x) \geq x^{0.84}$ of Krasikov–Lagarias [6]. Our pressure identity reproduces the exponent-1 statement of this conjecture as the $\alpha_- = 1$ case of a single closed-form equation, together with its $q \geq 5$ analogue; it does not resolve the conjecture. A calibrated empirical test of this conjecture at $q = 5$ against a competing prediction is the subject of a companion paper [2].

We state Theorem 5.1, Proposition 5.2, and Conjecture 5.3 as three separate claims, not one theorem about the arithmetic tree, because the step from one to the other is exactly the kind of transfer that Example 3.1 shows can fail; Remark 5.4 below specifies what does and does not rest on proof. Direct empirical support for Conjecture 5.3 is given in §7 below.

Remark 5.4 (What “transfers” rests on). For the i.i.d. branching-random-walk model that the annealed identity of Theorem 3.3 describes, quenched-equals-annealed growth throughout the unfrozen region is a rigorous, classical result [9]. For our arithmetic recursion, no analogous theorem is known to us: the closest arithmetic result in the literature is a one-sided density bound of Applegate–Lagarias [4, 5], later sharpened by Krasikov–Lagarias [6], not a matching two-sided quenched statement. We treat the transfer, at every root we invoke it for, as supported by (i) the classical stochastic-model theorem, (ii) direct numerical confirmation, and not by a proof for the arithmetic tree itself.

Conjecture 5.5 (Density and tail of the tilted Syracuse measure). *Let U be Haar-uniform on $\mathbb{Z}_q$, and read $Z_k(\alpha_-(q); U)$ through its residue modulo q^k . The sequence has the limit $W_q := W_{q, \alpha_-(q)}$ of Theorem 3.6. The tilted measure is absolutely continuous, its density W_q is non-degenerate, and*

$$\Pr(W_q > x) = x^{-\alpha_+(q)/\alpha_-(q)+o(1)}.$$

Here $\alpha_+(q)$ is the larger root of (4) ($\alpha_+(3) = 2$; $\alpha_+(q) = 1$ for every $q \geq 5$).

Theorem 5.6 (Tail index in the i.i.d. model). *For the additive martingale at parameter $\alpha_-(q)$ in the matching i.i.d. branching model,*

$$\Pr(W_\infty > x) \sim C_q x^{-\alpha_+(q)/\alpha_-(q)} \quad (x \rightarrow \infty)$$

for a constant $C_q > 0$.

Proof. Apply the implicit renewal theorem for generalized multiplicative cascades [15]. With $\theta = \alpha_-(q)$, its logarithmic moment function is

$$\psi(\sigma) = \log \rho_{\text{ann}}(\theta\sigma).$$

It vanishes at $\sigma = 1$ and next at $\kappa = \alpha_+(q)/\alpha_-(q)$, while $\psi'(1) < 0$ because the smaller root is unfrozen. The geometric offspring ladders give all moments needed in a neighborhood of κ . The closed subgroup generated by the logarithmic weights is generated by $\theta \log 2$ and $\theta \log q$; it is dense because $\log q / \log 2$ is irrational. Thus the renewal law is non-arithmetic and has a constant, positive tail coefficient. $\square$

The hypotheses of Liu’s theorem invoked above, namely $\rho_{\text{ann}}(\alpha_-(q)) = 1$, $\rho_{\text{ann}}(\alpha_+(q)) = 1$, $\psi'(1) < 0$, and $\kappa = \alpha_+(q)/\alpha_-(q) > 1$, were checked numerically for $q = 3, 5, 7, 9, 15$ in the accompanying repository.

This is the classical smoothing-transform result for the stochastic model. Its transfer to the coherent q -adic root environment is Conjecture 5.5, since the arithmetic subtrees are not independent. The tail-index equation $\mathbb{E}[\sum_i A_i^\sigma] = 1$ becomes, for our weights, $q^{\alpha-\sigma-1} = 2^{\alpha-\sigma} - 1$, solved by $\sigma = 1$

trivially and, nontrivially, by $\sigma = \alpha_+/\alpha_-$. This is further reason to take the conjecture seriously as a prediction, even though, per Remark 5.4, the theory rigorously covers only the stochastic model so far.

We state this as a conjecture separate from both Theorem 5.1 and Conjecture 5.3 above, because Proposition 4.1 shows $\alpha_+(q)$ is *always* frozen, so the annealed pressure identity alone does not establish the tail. Theorem 5.6 obtains it by implicit renewal at the unfrozen martingale normalization. Its arithmetic counterpart still requires evidence external to Theorem 3.3.

Every numerical test below samples actual integer roots of the arithmetic tree, not points of the Haar-uniform q -adic population U that Conjecture 5.5 is stated over. What these tests bear on directly is a different, equally open claim: the tail of the scale factor already introduced, conjecturally, in Proposition 5.2.

Conjecture 5.7 (Tail of the real-tree growth scale factor). *Fix odd $q \geq 3$. Let v range uniformly over the admissible roots in an interval growing to infinity, and set $W_v(H) := N_v(vH)/H^{\alpha_-(q)}$ for headroom $H \geq 1$ (N_v as in §2). As $H \rightarrow \infty$ and then the interval grows, $W_v(H)$ converges in law to the scale factor W of Proposition 5.2, with upper tail*

$$\Pr(W > x) = x^{-\alpha_+(q)/\alpha_-(q)+o(1)}.$$

The same implicit-renewal mechanism that proves Theorem 5.6 for the i.i.d. model predicts this exponent for W ,¹ exactly as it predicts it for W_q in Conjecture 5.5; the two conjectures are expected to agree numerically at every q , but neither is known to imply the other. W is indexed by an actual integer root, W_q by a Haar-random q -adic point, and no coupling between the two populations has been established.

At $q = 3$ the evidence for Conjecture 5.7 is real and threefold: the Hill estimator on real-tree samples and extreme-value theory on block maxima (both from earlier stages of this line of work) give $\alpha \approx 2$ independently of the pressure argument, and the direct enumeration of §7 (Empirical Result 7.1) adds a third, related line of support (that enumeration targets Conjecture 5.3, the growth exponent, a distinct claim from Conjecture 5.7, tested further below). For $q \geq 5$ the evidence for Conjecture 5.7 is markedly weaker: an early Hill-estimator measurement of finite-depth root scale factors for $q = 5$ (600 roots, agreement to two decimal places with the predicted 1.536...) was later shown, over the course of this line of work, to be statistically non-confirmatory at that sample size (the estimator's true standard error is ≈ 0.45 , an order of magnitude larger than the agreement being claimed). Two further tests bear on Conjecture 5.7 and Conjecture 5.5, one on each.

First, on Conjecture 5.7: a larger statistical test (5000 roots, four headroom levels up to 10^8) applied a battery of four estimators (a rank-size regression with the Gabaix–Ibragimov finite-sample

¹A related structural prediction of the multitype model, confirmed empirically for $q = 5$: conditioning on the root's residue class $i = u_0 \bmod q$ (with $a_0(i)$ the smallest admissible branch exponent for class i), W decomposes by class as $W_i \stackrel{d}{=} 2^{-a_0(i)\theta} \cdot W^*$ for a common W^* , i.e. residue class only rescales the candidate limit, it does not change its shape. Verified on 5000 sampled roots across four headroom levels, with quantile ratios matching this prediction to within 2–9%; exact only in the idealized i.i.d. model, as in Remark 5.4. The same relation survives unchanged when 2 is not a primitive root mod q (the extra sterile classes of §2): restricted to the surviving types $\langle 2 \rangle$, one can still form a multitype pressure matrix, indexed by residue class, whose entries are one-step transition weights rather than the deterministic map that Example 3.1 shows cannot exist; it is well defined because the digit $k = \lfloor u/q \rfloor \bmod q$ that example identifies as the missing information is itself uniform over all q residues independently of the parent's class i , so the transition weight out of class i does not depend on i . This pressure matrix is rank 1, and its Perron eigenvalue collapses, by a d -independent cancellation (a_0 a bijection $\langle 2 \rangle \rightarrow \{1, \dots, d\}$), to exactly $q^{\sigma-1}/(2^\sigma - 1)$, the same pressure equation, so both the exponent and the scale-family direction survive intact; verified for $q = 7, 15$ to 1–4%. (Whether 2 is a primitive root mod q is, for prime q , governed by Artin's primitive root conjecture, open in general but proved conditionally on GRH by [16].)

correction; a bias-corrected Hill estimator following Huisman et al.; a generalized-Pareto threshold-stability scan; and a Clauset–Shalizi–Newman fit with a Vuong test against a truncated log-normal alternative). The outcome is mixed rather than confirmatory: the bias-corrected Hill estimator is stable across headroom levels and lands close to the predicted 1.536, but the threshold-stability scan shows no clean plateau near the predicted shape parameter, and the Vuong test favors the log-normal alternative over the power law, at conventional significance, in three of the four headroom levels tested. Reading the four estimators together by the depth of tail they each summarize shows the apparent local tail index rising smoothly with depth (from ≈ 1.3 at the widest window to ≈ 2.2 at the narrowest) without settling near any single value: consistent with a slowly-converging pre-asymptotic regime in which no estimator at this sample size can be treated as decisive.

Second, on Conjecture 5.5 itself: we tested it by an exact, non-statistical route: the same depth- k recursion $Z_k(\theta; u)$ used to verify the frozen-phase growth law above (with $\theta = \alpha_-(5)$) was evaluated over the *entire* population of residues $u \bmod 5^k$ (not a sample), for k up to 11 ($5^{11} \approx 4.9 \times 10^7$ residues; a sanity check, that the population mean of $Z_k(\theta; u)$ equals exactly 1 for every k , forced by Theorem 3.3 at $\alpha = \theta$, passed to floating-point precision, confirming the implementation). If $W_q = \lim_k Z_k(\theta; U)$ has tail index ι , the population moment $M_k(p) = \text{mean}_u Z_k(\theta; u)^p$ should saturate in k for $p < \iota$ and diverge for $p > \iota$. The data show saturating behavior throughout $p \leq 1.6$ and diverging behavior for $p \geq 1.7$, which would place ι higher than the predicted 1.536 if taken at face value; the ratio of successive increments has not yet stabilized for $p \leq 1.6$ at $k = 11$ (unlike $p \geq 1.7$, where it has), the classic signature of a system still relaxing toward its asymptotic exponent rather than one that has reached it. Given the previously observed slow convergence rate for $q = 5$ ($k^{-0.222}$ decay in the moment test, not an effect of the linear transfer operator’s spectrum, which we later verified to be exactly $\{\Lambda, 0\}$ at every truncation level, i.e. a perfect spectral gap with no isolated subdominant eigenvalue; the transient’s origin lies in a separate, nonlinear critical layer and remains open), the transient decays by only about 7% between $k = 8$ and $k = 11$; halving it would require $k \approx 250$, far beyond what exhaustive enumeration over 5^k residues can reach. We therefore cannot distinguish, from this data, between $\theta = 1.536$ lying just below a nearby true index and lying further below one. The test is inconclusive, not disconfirming, and enumeration alone will not resolve it.

Neither route confirms its target for $q \geq 5$ (Conjecture 5.7 for the first, Conjecture 5.5 for the second), and neither refutes it; both are more informative than the original under-powered measurement, and both point to the same conclusion: this is a harder statistical question than the $q = 3$ case, which has stronger finite-level support from the independent Hill/EVT measurements noted above. Two follow-ups address the two open questions separately. The first targets the unresolved transient in the exact test of Conjecture 5.5: a log-periodic fit (motivated by the branch weights being powers of 2) was tested against a period derived analytically, not fitted: the relevant renewal-theoretic dichotomy is non-arithmetic (the shift $\log_2 5$ is irrational), so no asymptotic log-periodicity is predicted, and a periodogram test found no power at the predicted frequencies. An analytic handle on the linear transfer operator’s spectrum, once obtained, likewise turned out not to bear on this question, since the transient lives in a separate nonlinear layer (the operator has a provably perfect spectral gap). The second follow-up extends the root battery behind Conjecture 5.7: a comparison on a $20\times$ larger sample (10^5 roots) produced, for the first time, a clean GPD threshold-stability plateau at the predicted value, a tightly concentrated Hill estimate, and a Vuong test that no longer favors the lognormal alternative; both were checked against two synthetic calibration tests. This is now the strongest evidence to date in favor of Conjecture 5.7 for $q = 5$, though not a proof of it: Vuong non-rejection is not confirmation, and $W_v(H)$ ’s own median drifts from 2.04 at $H = 10^5$ to 0.94 at $H = 10^8$, direct evidence the scale factor has not converged at the headroom reached.

The mean drift $\log q - 2 \log 2$ changes sign exactly at the threshold $q = 5$ separating the two cases of Conjecture 5.3: negative for $q = 3$ (orbits contract on average), positive for $q \geq 5$ (orbits expand on average). The same sign condition governs the freezing computation of §4: at $\alpha = 1$ the entropy is exactly $s(1) = \log(4/q)$, so the same sign that decides contraction versus expansion of orbits also decides whether the trivial root $\alpha = 1$ is frozen.

6 The tail-exponent regime at $q = 3$

At $q = 3$, the second root $\alpha^* = 2$ is numerically the boundary value that the classical multivariate smoothing-transform theory [17] would classify as the *critical case*, characterized by $m'(\alpha) = 0$ at a root of $m(\alpha) = 1$ (a boundary/degenerate case in the sense of Biggins–Kyprianou). Writing $m(\alpha) := q^{\alpha-1}/(2^\alpha - 1)$ (the pressure function of §3 at $q = 3$), we compute

$$m'(1) = \ln 3 - \ln 2 \cdot \frac{2^1}{2^1 - 1} \quad \text{explicitly} \approx -0.288 < 0, \quad m'(2) = \ln 3 - \frac{4}{3} \ln 2 \approx +0.174 > 0.$$

By Lemma 3.5, $\log m$ is strictly convex, so the two distinct roots of $m(\alpha) = 1$ necessarily straddle its unique minimum, so *neither* can be a critical (double) root; criticality would require the two roots to merge, which does not happen at $q = 3$. The correct classification of $\alpha^* = 2$ is therefore as the *second root with $m' > 0$* : a Goldie/Guivarc’h-type regularly-varying tail regime [11, 18], not the critical case. This distinction matters operationally: the critical-case uniqueness theorem of [17] *presupposes* exactly the independence between subtree copies that the endogeny-barrier companion paper identifies as the arithmetic gap obstructing a full proof [1]. Even setting aside the regime mismatch, it could not have supplied a new tool for closing that gap. Tying this to §4: “second root with $m' > 0$ ” is exactly the freezing condition of Proposition 4.1: the Goldie/Guivarc’h regime name correctly classifies the algebraic root $\alpha^* = 2$ of $m(\alpha) = 1$, but by itself says nothing about whether this root governs the *quenched* tail of our specific arithmetic recursion. That is exactly the question Proposition 4.1, Conjecture 5.5, and Conjecture 5.7 address directly.

7 Empirical confirmation on real reverse trees

Empirical Result 7.1 (Direct enumeration on real reverse trees). *Independent depth-first enumeration of the reverse trees for $q = 5$ and $q = 7$, following the exact admissibility rule of §2, provides direct empirical support for Conjecture 5.3: counting slopes per decade of 3 independently sampled roots gave ≈ 0.62 – 0.66 for $q = 5$ (predicted 0.650919) and ≈ 0.36 – 0.39 for $q = 7$ (predicted 0.373501).*

A more careful, statistically calibrated measurement for $q = 5$, separating this prediction from a competing one, is the subject of a companion paper [2]. As already noted above for $q = 3$ (where $w_2(1) = 1$ is a fixed point), the smallest-branch map $u \mapsto w_{a_0(u \bmod q)}(u)$ has finite cycles for $q = 5$ (e.g. $1 \mapsto 3 \mapsto 1$, since $w_4(1) = 3$ and $w_1(3) = 1$) and $q = 7$; naive enumeration would loop on nodes that reach one. Our implementation excludes all such cycle members explicitly (precomputed for each q) and deduplicates visited nodes, instead of only sampling roots far from them.

Remark 7.2 (Bibliographic calibration). Neither the numerical value $\alpha^*(5) = 0.650919$ nor the qualitative transition at $q \geq 5$ is new, and a later result rigorously excludes $q \geq 5$ in the forward direction for the analogous $p = 2$ generalization [8]. Wirsching [7, Ch. III, §5, following Theorem 5.2] anticipated the same qualitative transition at $p > 4$ heuristically in 1998, in his own

terms (a sign flip in the index of the maximal term $f_p(n)$, leading him to expect vanishing asymptotic density for the weak components of the Collatz graph of T_p), and Kontorovich–Lagarias [3] had already computed the same quantity (denoted $\eta_{5,BP}$ in their Theorem 8.10) via a different route, large-deviations methods for a branching random walk, that is numerically identical to our pressure-equation derivation. Given §4, this difference in route matters beyond giving an alternative derivation: a large-deviations argument for a branching random walk is, in the appropriate sense, a quenched statement, whereas our pressure identity (Theorem 3.3) is annealed. That the two agree numerically at $\alpha_-(5)$ is consistent with $\alpha_-(5)$ being unfrozen (Proposition 4.1), where quenched and annealed are expected to coincide; it does not by itself certify that our annealed route would agree with a quenched one at a frozen root. The closed-form equation (4) is new: it unifies the family for arbitrary q in a single algebraic identity, with a rigorous annealed derivation and independent numerical confirmation.

Data Availability Statement

Code and data reproducing every claim in this paper, with a README mapping each script to the section and result it verifies, are at <https://github.com/faculdade/collatz-qx1-pressure>.

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